

CardioFusionNet: 1D-CNN, Isolation Forest-XGBoost Gated ECG Arrhythmia and ECG-PCG Anomaly Detection

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Abstract — Cardiovascular diseases are a major cause of death worldwide. This paper introduces two related systems for analysing cardiac signals automatically. The first system uses 1D CNN to classify arrhythmias based on ECG signals from the MIT-BIH Arrhythmia Database. It applies beat-level segmentation, Butterworth bandpass filtering, and AAMI-standard class mapping. The trained model reaches an overall accuracy of 97% and an average F1 score of 0.81. The second system suggests a multimodal anomaly detection pipeline. It combines statistical features from synchronized (ECG) and phonocardiogram (PCG) signal windows. It uses Isolation Forest to generate unsupervised pseudo-labels. Then, a tuned XGBoost classifier with probability thresholds achieves 98.24% accuracy and a binary F1 score of 94.22%. A window-based cardiac risk scoring method is proposed to help with continuous patient monitoring. These systems show that both deep learning and machine learning methods can effectively analyze cardiac signals. They have real-world applications for detecting arrhythmias in real time and for monitoring heart health using multiple modalities.

Keywords — Cardiovascular diseases; beat level segmentation; Butterworth bandpass filtering; phonocardiogram; unsupervised pseudo-labels; arrhythmias.

1. INTRODUCTION

The rapid progress of sensor technologies, wireless communication, and machine learning (ML) algorithms has enabled AI (Artificial Intelligence) driven Remote health monitoring (RHM) solutions that enable real-time monitoring of physiological data and early detection of diseases. The amalgamation of AI and Internet of things (IoT) in the healthcare has immensely revolutionized the patient monitoring and disease care. The rapid progress of sensor technologies, wireless communication, ML algorithms has enabled AI RHM solutions that enable real-time monitoring of physiological data and early detection of diseases. With billions of devices connected it is more crucial than ever to effectively manage these devices for long term chronic care conditions [1], [2], [3]. They gather heart rate, blood pressure and levels of oxygen in the blood and can obtain electrocardiogram (ECG) readings, thus allowing healthcare providers to track patients remotely and minimising hospital visits and increasing access to immediate healthcare interventions AI analytics have the ability to find irregularities in health data, forecast potential medical issues and inform personalized treatments [2], [3].

The healthcare sector, over the last few years, has been under pressure from the population increase, the rise of chronic diseases, and shortage of medical resources. A traditional healthcare system is mainly based on check-ups conducted regularly and treatment that is given only when symptoms appear or are already at a severe stage. Through this approach, in many cases, diagnosis is delayed, and the possibility of the effective treatment is low [4], [5], [6]. Thus, the necessity of a new system that would be able to constantly monitor patients and detect health risks in the earliest stage is very strong.

In recent years, the accelerating trend of global population ageing has led to a growing demand for health management among older adults, thereby driving the rapid development of wearable devices within this demographic. Equipped with multiple sensors and intelligent algorithms, wearable devices can collect, monitor, and analyse older adults physiological and behavioural data in real time [7], [8], [9]. Thereby supporting health management, disease prevention, and rehabilitation training. Owing to their portability, functionality, and diverse features, wearable devices have gradually become an important technological tool for elderly health management.

The application of AI in the deployment of wearable sensors is an indispensable part of the revolution of the IoTs. Consequently, the rapid development of AI techniques has provided health professionals with a greater capacity to deal with the huge amounts of data collected through wearable devices that have been deployed in monitoring patients' health condition [10].

AI improves the ability to explore the relationships between the information acquired from the sensor signals' output and the health status of individuals through the establishment of various types of diagnostic and predictive models. [11]. Understanding AI applications in wearable sensor data manipulation is, therefore, of paramount importance in optimising the diagnosis and prediction of Cardiovascular disease (CVD).

Older adults have distinct and complex needs for precision health management because of the pathological and physiological changes that occur with aging. As people age, their physiological and biochemical indicators undergo specific changes, and even subtle variations in these changes can signal the onset of disease, making the sensitive detection and interpretation of these signals essential [10], [11], [12]. Consequently, while wearable devices enable the collection of vast amounts of health data, researchers have devoted considerable effort to enhancing the quality of this data. Biophysical and biochemical information, such as heart rate, blood pressure, glucose levels, and biomarkers, can be captured by these devices and used for diagnosis and intervention when cardiovascular conditions, diabetes, and other health disorders are present.

AI biosensors are equipped with advanced algorithms that can process data directly from the sensor to provide actionable insights, detect anomalies, and predict health events in real time. This paradigm shift has the potential to not only improve the accuracy and efficiency of health monitoring but also contribute to personalized medicine by adapting to individual needs and preferences. With the increasing availability of data and the rise of machine learning techniques, AI has become an indispensable tool in enhancing the functionality and usability of wearable sensors. However, the application of AI in wearable sensors presents unique challenges [13], [14], [15]. These include ensuring model robustness in the presence of distribution shifts, where sensor data can vary across different environments and populations, and developing personalized models that can adapt to individual users over time. Furthermore, the integration of edge AI and systems adds another layer of complexity, requiring seamless interaction between the device and its user to refine predictions based on user feedback [16].

ECG, which records the electrical activity of the heart over time, serves as the standard diagnostic tool for arrhythmia identification, while the phonocardiogram (PCG), which captures the acoustic signals produced by heart valve movements and blood flow, provides complementary information regarding the mechanical function of the heart. However, manual analysis of these signals is inherently time consuming, and difficult to scale to the demands of continuous or large cardiac monitoring [17], [18], [19]. These limitations have motivated growing interest in the development of automated, ML approaches capable of reliably analyzing cardiac signals with minimal human intervention.

Key Contribution

- A supervised deep learning approach 1D-CNN for fine grained beat level arrhythmia classification on ECG alone, and an unsupervised and semi-supervised approach for coarser anomaly detection on fused ECG-PCG signals, demonstrating how detection strategy should adapt to signal modality and label availability.
- Rather than treating ECG and PCG as 2 diagnostic streams, it combines them into a single joint representation, capturing information about electromechanical cardiac function that neither signal alone reveals.
- The 1D-CNN pipeline explicitly confronts the real world of AAMI beat using SMOTE, rather than inflated accuracy on an imbalanced test distribution, reflecting the class distribution clinicians actually encounter in ambulatory monitoring.

Section Organization

The paper starts with Abstract, a summary of both systems. Section 1 talks about Need for automated cardiac monitoring covers the AI & IoT healthcare, the limitations of manual ECG&PCG interpretation. Section 2 contains four subsections ECG&PCG analysis gap, hybrid deep learning and classical ML, semi-supervised labelling via Isolation Forest, threshold optimization for imbalanced classification. Section 3 contains datasets MIT-BIH and Phonogram, 1D CNN pipeline, multimodal pipeline, Key Contribution and Section Organization. Section 4 talks about results and discussion where results are discussed and a discussion comparing others papers including state of the art comparison. The paper ends with conclusion, future work if needed and all the papers that have been used for reference.

2. RELATED WORK

Several recent studies have applied AI and ML to wearable and ECG based health monitoring. Most of all comparably are being the noticeable are being mentioned as follows, Sharma et al. [1] proposed an AI-driven health monitoring system that collects heart rate, blood pressure, temperature, and SpO₂ from wearable IoT sensors and applies ML algorithms such as SVM and ANN to detect abnormal patterns, reporting 92–95% accuracy in identifying at-risk conditions. Similarly, Ramzan et al. [2] developed an IoT-based remote health monitoring system and compared Random Forest, SVM, and KNN classifiers on physiological data, finding that Random Forest consistently outperformed the other models (96.8% testing accuracy, ROC-AUC of 0.97). Zhi et al. [3] conducted a bibliometric analysis of a decade (2014–2024) of research on wearable devices for elderly health management, showing that the field has grown at an annual rate of 7.65% and that research focus has shifted from early fall-detection and activity-tracking work toward heart rate variability, atrial fibrillation, and AI-driven analytics. Soumma et al. [4] and J D Huang et al. [5] further review the application of AI to wearable sensor data for cardiovascular monitoring and diagnosis. Z.Li et al. [6] explores the integration of nano sensor enhanced wearables with machine learning for precision health management in older adults, while Z.Yan, J.Zhoe et al. [7] proposes an energy-efficient spiking neural network for ECG classification.

At the signal level, Lai et al. [8] evaluated four non standardized patch-based ECG lead positions together with CNN based classifiers for atrial fibrillation detection, finding that the modified lead II position (MP1) combined with an R R interval based CNN (RRI CNN) gave the best performance (93.1% accuracy, 93.1% sensitivity, 93.4% specificity) and outperformed algorithms trained without accounting for lead placement variability. Pan et al. [9] proposed MCA net, a multi task channel attention network that processes all 12 ECG leads independently before fusing them, achieving over 90% accuracy for myocardial infarction detection and outperforming prior single-task CNN and ResNet-based models on both the PTB and PTBXL datasets. Li et al. [10] took a multi-modal approach, combining deep-coded ECG and PCG features selected via a genetic algorithm and classified with an SVM, and showed that this fusion strategy (AUC of 0.936) outperformed either single-modality model alone (ECG-only AUC 0.916; PCG-only AUC 0.748).

Author(s)	Databases	Data Signals	Algorithms	Evaluation
N.Sharma et al. [1]	Sample data	ECG	ANN -SVM	ACC: 92-95 %
Ramzan et al.[2]	Publicly available datasets	ECG	Random forest, SVM, KNN Classifier	ACC: 96.8%
Zhi et al. [3]	Web of Science Core Collection	N/A	Bibliometric mapping	80.8%
Soumma et al. [4]	Sample data	ECG	Edge-AI	N/A
Huang et al. [5]	Review	N/A	ML/DL Approaches	N/A
Li et al. [6]	Review	N/A	EDGE-AI	N/A
Z. Yan, J. Zhou et al. [7]	MIT BIH	ECG	SNNs	ACC:90%
D. Lai, Y. Bu et al. [8]	MIT BIH	ECG	CNNs	ACC:93.1
W. Pan, Y. An et al. [9]	PTB and PTBXL	ECG	CNNs	ACC: 88.59
P. Li, Y. Hu [10]	PhysioNet/CinC	ECG + PCG	GA-SVM	ACC: 87.40

Table 1. Relative comparison with other studies.

While recent work spans a wide spectrum from broad AI health monitoring systems to cardiac models most lack a standardized, public dataset or address only a single narrow task. Studies such as [1] and [2] report strong accuracies (92–95% and 96.8%) but rely on vitals data without a named or explicit imbalance handling, while [3]–[6] are review bibliometric works that survey trends rather than propose or evaluate a model. Among cardia specific studies, [7] optimizes an SNN on MIT-BIH primarily for energy efficiency rather than full arrhythmia detection performance, [8] and [9] target single conditions using single or ECG alone, and [10], the closest precedent to our multimodal pipeline fuses ECG and PCG via deep coding features and a optimized SVM but does not validate against verified ground truth. In contrast, this study benchmarks a 1D-CNN on the full inter-patient MIT-BIH split with explicit SMOTE based class imbalance correction and macro F1 Score, and extends this with a 2 ECG–PCG framework combining unsupervised anomaly detection (Isolation Forest) and supervised classification (XGBoost) while transparently acknowledging that our "ground truth" labels are Isolation Forest pseudo labels, a limitation rarely made.

3. METHODS AND MATERIALS USED

This paper addresses both challenges through two interconnected contributions. 1D CNN is developed for the classification of cardiac arrhythmias from ECG signals sourced from the MIT BIH Arrhythmia Database, a widely adopted benchmark in the cardiac signal processing literature. The pipeline incorporates beat-level segmentation on R peaks, Butterworth bandpass filtering for noise removal, and mapping to the four standardized AAMI beat classes: Normal (N), Supraventricular ectopic (S), Ventricular ectopic (V), and Fusion (F). The class imbalance to real world arrhythmia datasets is addressed through class weighting during model training.

CNNs, have strong performance to 1D physiological time-series data, effectively capturing local patterns and features within cardiac waveforms without requiring manual feature engineering. Classical ML approaches such as gradient boost decision trees have proven highly effective for structured, feature engineered representations of physiological signals, particularly where labelled training data is limited or must be generated through unsupervised methods..

A multimodal pipeline is proposed that moves single modality analysis by fusing statistical features extracted from ECG and PCG signal windows. An Isolation Forest model is made to generate pseudo-labels through unsupervised anomaly scoring, with a percentile based threshold to reflect an anomaly assumption. These pseudo labels are subsequently used to train a tuned XGBoost classifier, with probability threshold optimization applied post-training to maximize F1 performance.

Together, these two systems show dimensions of cardiac monitoring: the first shows fine grained arrhythmia classification from ECG , while the second shows the added value of multimodal signal fusion for cardiac anomaly detection in a supervised setting. Both shows the practicality of machine learning for real time cardiac health monitoring, and suggest that deep learning based waveform analysis with classical feature driven approaches.

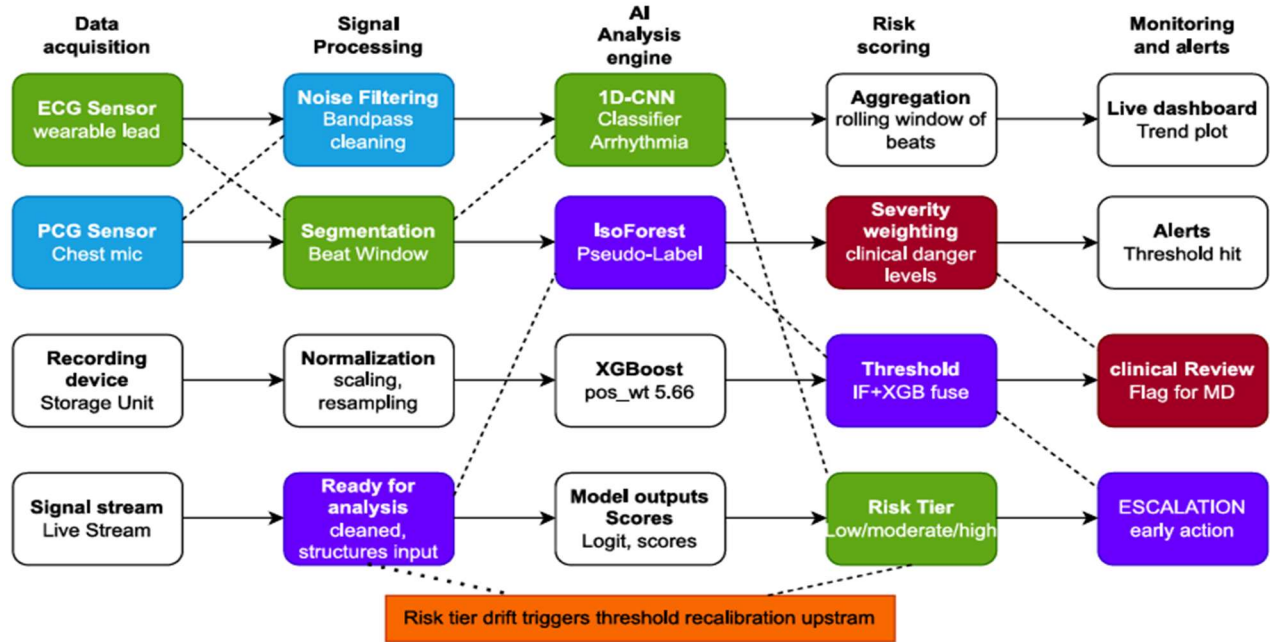


Figure 1: Cardiac Risk Monitoring System

Figure 1 shows an architectural summary of how this Multi Modal AI Based Cardiac risk Monitoring system has been built.

3.1. DATASETS

MIT-BIH Arrhythmia Database (System 1):

The MIT BIH (Massachusetts Institute of Technology and Beth Israel Hospital) Arrhythmia Database is one of the most widely adopted benchmarks in ECG based arrhythmia research, developed jointly by MIT and Boston's Beth Israel Hospital to support reproducible evaluation of automated rhythm classification algorithms. The dataset comprises 48 half hour excerpts of two channel ambulatory ECG recordings collected from 47 subjects, digitized at 360 Hz per channel with 11-bit resolution over a 10 mV range, providing sufficient temporal and amplitude precision to capture subtle morphological variations in the QRS complex. Each recording includes a modified limb lead II alongside a second lead (typically V1, V2, V4, or V5 depending on the subject), enabling analysis of beat morphology from two complementary electrical perspectives and improving the robustness of beat detection across different cardiac axis orientations.

Every beat in the database has been independently annotated by two or more cardiologists, with disagreements resolved through consensus, yielding over 109,000 beat-level labels that serve as reliable ground truth. For this study, individual beat annotations were mapped to the four AAMI recommended super classes Normal, Supraventricular ectopic, Ventricular ectopic, and Fusion a grouping that reflects standard clinical practice and simultaneously introduces pronounced class imbalance, since normal beats vastly outnumber ectopic and fusion beats in real-world monitoring conditions

Ephnogram Database (System 2):

The Ephnogram database is a purpose-built multimodal resource designed to support joint electrical and acoustic analysis of cardiac activity, addressing a gap left by single-modality ECG databases such as MIT-BIH. It consists of simultaneously recorded ECG and PCG signals acquired from the same subjects under matched conditions, allowing direct temporal correspondence between the heart's electrical depolarization-repolarization cycle and the mechanical sounds produced by valve closure and turbulent blood flow. This simultaneity is central to the dataset's value, since it permits analysis of electromechanical coupling.

This combined representation was then used as input to an unsupervised anomaly detection stage, reflecting the underlying premise that abnormalities affecting cardiac electrical and mechanical function often manifest jointly, and that a fused representation can surface anomalies that might be missed when either signal type is analysed independently.

3.2. Proposed Architecture

The 1D CNN Arrhythmia Classification is designed to process the segmented ECG beat waveforms. It consists of three stacked Conv1D, followed by ReLU activation for non-linearity and Batch Normalization for training stability. The pooling layers decrease the resolution but keep the important features. The resulting feature maps are passed to a Dense layer and subsequent Dropout for regularization. The last Softmax layer gives probabilities for the four AAMI classes (N, S, V, F). The model was trained with class weighting to handle imbalance, EarlyStopping and a batch size of 512 using the Adam optimizer. For multimodal hybrid model is a two stage multimodal anomaly detection pipeline combining ECG and PCG signals. Raw ECG and PCG signals are first preprocessed and segmented, then features are extracted from both modalities to form a combined feature representation. In the first stage, an unsupervised Isolation Forest model is applied to this feature set to identify anomalous segments, generating pseudo labels. In the second stage, an XGBoost classifier is trained using these Isolation Forest outputs as target.

3.2.1. 1D-CNN Arrhythmia Classification

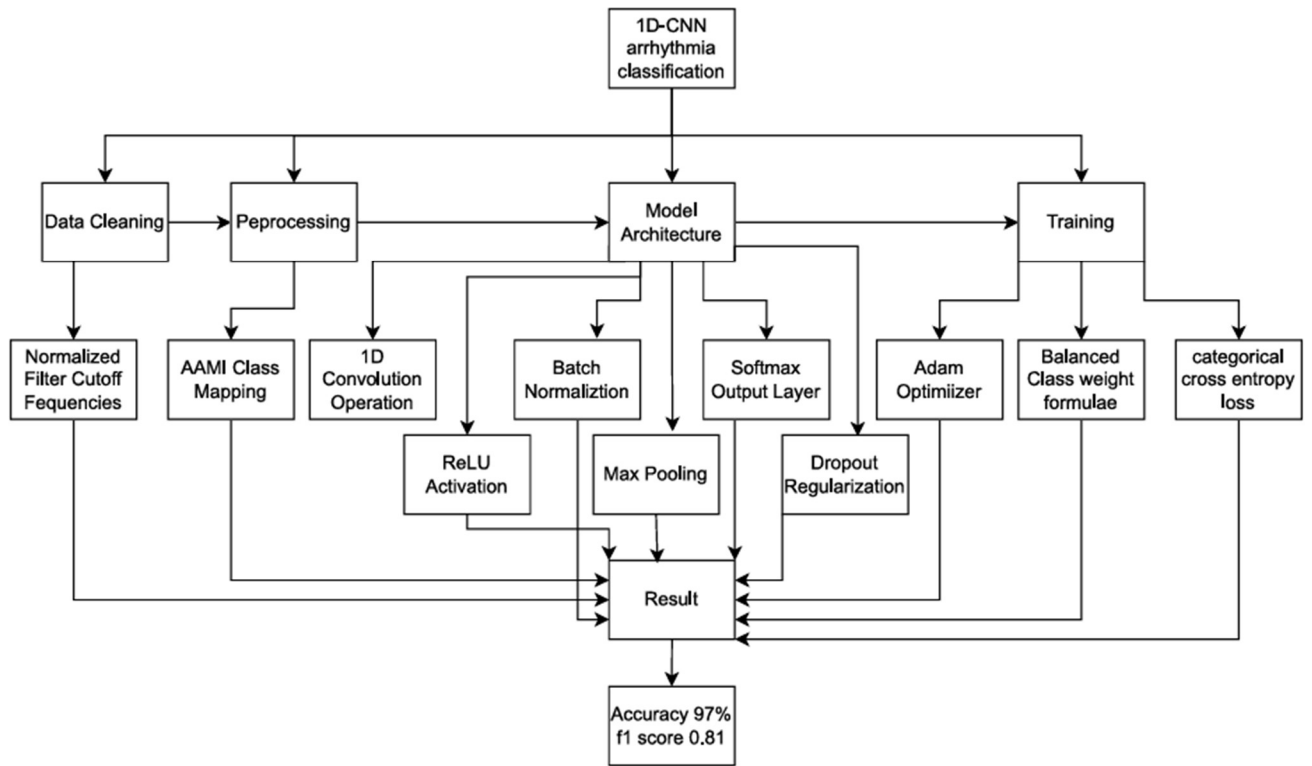


Figure 2: 1D CNN Arrhythmia Classification

The raw data from the MIT BIH Arrhythmia Database after preprocessing of bandpass filtering with a Butterworth filter and beat segmentation around the R peak result in 101,024 beat segments classified into four AAMI categories. split into training and test sets is used to train a three layered one dimensional CNN model consisting of the layers Conv1D, Batch Normalization, MaxPooling, and then dense with dropout, with class weights and early stopping to

combat imbalanced classes. Evaluation of the trained model is done through accuracy, macro F1 score, and confusion matrix with further calculation of a weighted cardiac risk score based on windows.

3.2.1.1. Data Cleaning

Second Order Butterworth Bandpass Filter

For filtering the noisy signals and in order to get a clean filtered signals from the ECG and PCG the second order butter worth bandpass filter has been applied .The function of the Butterworth filter of order n is defined as:

$$| H(j\Omega) |^2 = 1 + \frac{1}{1 + (\frac{\Omega}{\Omega_c})^{2n}} \quad (1)$$

where:

- $H(j\Omega)$ is the frequency response.
- Ω is the input angular frequency.
- Ω_c is the cutoff angular frequency.
- $n=2$ is the filter order, providing smooth roll-off without passband.

The filter is applied using zero phase forward backward filtering (filtfilt), which processes the signal in both forward and reverse directions, ensuring output has zero phase distortion a critical property for preserving the morphological shape of ECG waveforms that are used for clinical beat classification.

Zero phase filtering: Applied forward and backward through signal. Prevents phase distortion, which is critical so ECG waveform shape isn't changed by the filter.

Invalid label removal

Only beats with 'N', 'L', 'R', 'A', 'V', 'F', 'J', 'E' are kept, dropping unclassifiable beats.

Boundary cleaning

Beats too close to the recording are excluded, since a full window can't be extracted it prevents noisy beat segments from entering the dataset.

3.2.1.2. Signal Processing

Normalized Filter Cut-off Frequencies

The frequencies as which are essential has been either high or low and hence, in order to balance it filter cut-off frequencies which are normalized in order to get a proper processed signals for further usage that is indeed required for algorithm usage and for more and more evaluation metrics the normalized filter cut-off frequencies has been used. The formulas have been given as below along with the terms of what each means.

$$f_{low} = \frac{f_{lowcut}}{f_{nyq}}, f_{high} = \frac{f_{highcut}}{f_{nyq}} \quad (2)$$

where:

$f_{lowcut} = 0.5\text{Hz}$ is the lower cutoff frequency, chosen to suppress baseline wander slow drift in the ECG signal caused by patient respiration.

$f_{highcut} = 40$ Hz is the upper cutoff frequency, chosen to suppress high frequency noise including electromyographic (muscle) artifacts and power line interference.

f_{nyq} is the Nyquist frequency computed above.

Normalizing the cutoff frequencies to the range $[0,1]$, $[0, 1]$, $[0,1]$ relative to the Nyquist frequency is required by SciPy's Butterworth filter design function, which expects dimensionless frequency inputs rather than absolute Hz values.

3.2.1.3. Data Preprocessing

AAMI Class Mapping

The MIT BIH beat annotations were divided into four clinically standardized AAMI classes $N(\text{Normal})=\{N,L,R\}$, $S(\text{Supraventricular})=\{A,J\}$, $V(\text{Ventricular})=\{V,E\}$, $F(\text{Fusion})=\{F\}$. Beats falling outside these categories (Q class) were excluded from the dataset. This mapping follows the AAMI standard for arrhythmia classification, reducing the original MIT BIH annotation symbols into 4 clinically meaningful categories suitable for supervised learning.

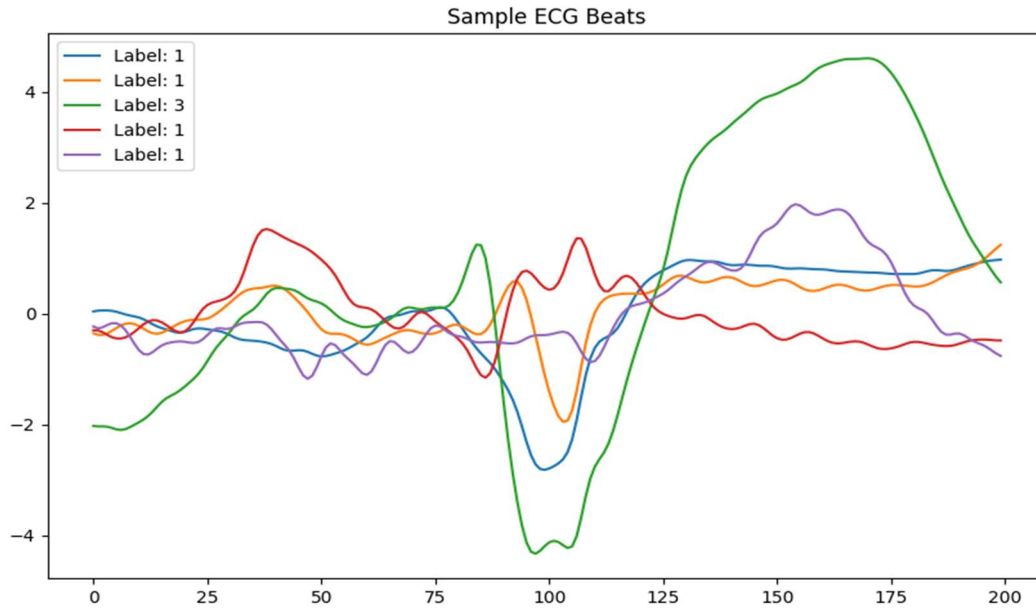


Figure 3: ECG beat segments

Figure 3 shows heart beat patterns from the MIT BIH Arrhythmia Database after cleaning. The signals were made clearer by using Butterworth bandpass filter then cutting them up around the R peaks. Made them all the same length with 200 samples. The heartbeat patterns can be seen that they are all a little different in terms of shape and size and that is what the proposed 1D CNN uses to tell apart the types of arrhythmia. The heartbeat patterns are really different, from one another which is what the 1D CNN looks at to classify the different arrhythmia classes.

3.2.1.4. Training

Balanced Class Weight Formula

In order to balance the weightage of each classes for the AAMI Class Mapping, a balancing weightage of each class so as to get satisfactory results ,the prerequisite need for balancing each classes with respect to weightage is essential in order to get the needed results for evaluation. The formulae for balanced class weight formulae along with what each term means and is defined has been given below.

$$w_c = \frac{N}{K \times N_c} \quad (3)$$

where w_c is the weight assigned to class c, N is the total number of training samples, K is the number of classes (K = 4 for the AAMI classes N, S, V, F), and N_c is the number of samples belonging to class c. This formula, used by scikit-learn's compute class weight, assigns larger weights to underrepresented classes (such as Supraventricular and Fusion beats) and smaller weights to the dominant Normal class, ensuring the loss function removes misclassification of rare arrhythmia types more heavily during training.

Sparse Categorical Cross-Entropy Loss

The sparse Categorical Cross entropy, is essential for training the balanced data after each class has been balanced with respect to the balanced class weight formulae, It is the standard loss for multi class classification like the ECG arrhythmia CNN when labels are integers that is 0,1,2,3 rather than one hot encoded vectors.

$$L = -\frac{1}{N} \sum_{i=1}^N \log(\widehat{y_{i,y_i}}) \quad (4)$$

where N is the number of training samples, y_i is the true class index for sample i, and $\widehat{y_{i,y_i}}$ is the predicted probability assigned to the true class. This loss function is minimized during training using the Adam optimizer, with per-class weighting applied via the balanced class weights defined above.

Adam Optimizer

Model parameters are updated using the Adam (Adaptive Moment Estimation) optimizer, which maintains per parameter adaptive learning rates based on first and second moment estimates of the gradients:

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1) g_t \quad (5)$$

$$v_t = \beta_2 v_{t-1} + (1 - \beta_2) g_t^2 \quad (6)$$

$$\widehat{m}_t = \frac{m_t}{1 - \beta_1^t}, \quad \widehat{v}_t = \frac{v_t}{1 - \beta_2^t} \quad (7)$$

$$\theta_{t+1} = \theta_t - \frac{\eta}{\sqrt{\widehat{v}_t} + \epsilon} \widehat{m}_t \quad (8)$$

where:

- g_t is the gradient of the loss with respect to parameter θ at step t.
- m_t is the first moment estimate (mean of gradients).
- v_t is the second moment estimate (uncentered variance of gradients).
- $\widehat{m}_t$ and $\widehat{v}_t$ are bias-corrected estimates to compensate for initialization at zero.
- $\eta = 0.001$ is the learning rate.

- $\beta_1=0.9, \beta_2=0.999$ are exponential decay rates.
- $\epsilon=10^{-7}$ prevents division by zero.

3.2.1.5. Architecture

1D Convolution Operation

By sliding a small filter that is kernel along the time axis of the ECG Signal, computing a weighted sum at each position to detect local patterns like QRS Complexes. It enables the network to learn local morphological features. The convolution operation computes a weighted sum of a sliding translating invariant patterns critical for arrhythmia detection

$$y[i] = \sum_{k=0}^{K-1} w[k] \cdot x[i + k] + b \quad (9)$$

where x is the input signal segment, w is the convolutional kernel of size K , b is the bias term, and $y[i]$ is the resulting feature value at position i . Three convolutional layers with increasing filter counts (32, 64, 128) and kernel sizes (5, 5, 3) were used to progressively extract low level to high level features from each 200-sample ECG beat window.

ReLU Activation

To introduce non linearity into the network, ReLU function is applied after each convolutional layer, it allows the model to lean complex, non linear relationships in the ECG Signal while mitigating the vanishing gradient problem common in deeper architectures.

$$f(x) = \max(0, x) \quad (10)$$

ReLU was applied after every convolutional layer to introduce non linearity while avoiding the gradient problem associated with sigmoid or tan activations, which is important for training deeper convolutional stacks on ECG beat.

Batch Normalization

Batch normalization is applied after each convolutional layer to stabilize and accelerate training. It normalizes the activation of each mini batch to have zero mean and unit variance, reducing internal covariate shift and allowing the use of higher learning rates while improving overall model.

$$\hat{x} = \frac{x - \mu_B}{\sqrt{\sigma_B^2 + \epsilon}}, \quad y = \gamma \hat{x} + \beta \quad (11)$$

where μ_B and σ_B^2 are the mean and variance of the current batch, ϵ is a small constant preventing division by zero, and γ and β are learnable scale and shift parameters. Batch normalization was applied after each convolutional layer to stabilize and accelerate training by reducing internal shift.

Max Pooling

After convolutional blocks to progressively reduce the temporal dimensionality of the feature maps max pooling is introduced. The network achieves translation invariance and reduces computational complexity while preserving the most salient feature of the ECG waveform.

$$y[i] = \max(x[2i], x[2i + 1]) \quad (12)$$

Max pooling with a pool size of 2 was applied after each convolutional block to progressively down sample the feature maps, reducing dimensionality while retaining most salient features of each beat.

Dropout Regularization

During training, each unit in the dense layer is retained with probability $p = 0.5$, and dropped otherwise, forcing the network to learn representations and reducing overfitting on the majority Normal beat class.

SoftMax Output Layer

SoftMax output layer converts the raw output logits into normalized class probabilities that sum to one, enabling the model to output a confidence score for each class and supporting multi class classification via the sparse categorical cross entropy loss.

$$P(y = j|x) = \frac{e^{z_j}}{\sum_{k=1}^K e^{z_k}} \quad (13)$$

where z_j is the output for class j , and $K = 4$ is the number of AAMI classes. The softmax function converts the final dense layer's raw outputs into a probability distribution over the four arrhythmia classes.

3.2.1.6. Downstream Application

Window-Based Weighted Cardiac Risk Score

Predicted beat labels are aggregated over rolling windows of 500 consecutive beats to produce a continuous cardiac risk score:

$$R_w = \frac{p_f^{(w)} \cdot w_f + p_s^{(w)} \cdot w_s + p_v^{(w)} \cdot w_v}{100} \quad (14)$$

where:

- R_w is the risk score for monitoring window.
- $p_f^{(w)}, p_s^{(w)}, p_v^{(w)}$ are the predicted percentages of Fusion, Supraventricular, and Ventricular beats within window w .
- $w_f = 2, w_s = 1, w_v = 3$ are clinical severity weights, reflecting the relative danger of each arrhythmia class.

The risk level is then classified as :

$$Risk\ Level = \begin{cases} Low\ Risk, & \text{if } R_w < 0.05 \\ Moderate\ Risk, & \text{if } 0.05 \leq R_w < 0.15 \\ High\ Risk, & \text{if } R_w \geq 0.15 \end{cases}$$

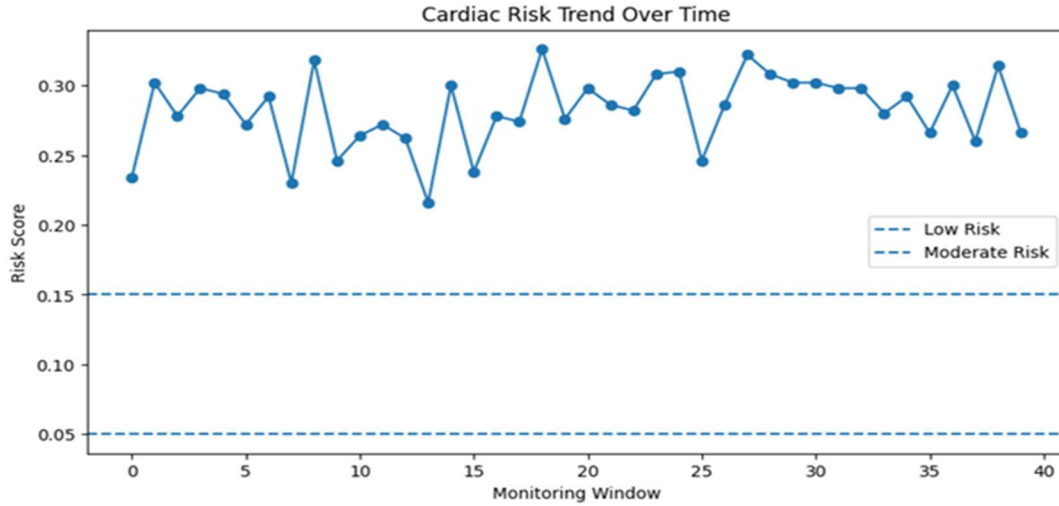


Figure 4: Cardiac Risk Trend

- X-axis: Monitoring Window (0–39), each representing a consecutive block of 500 beats from the test set predictions
- Y-axis: the weighted risk score R_w calculated for that window
- Dashed lines: the two thresholds — 0.05 (Low Risk cutoff) and 0.15 (Moderate Risk cutoff)

The proposed Figure 4 window-based risk scoring method provides a continuous cardiac risk score for each monitoring window. The trend shows the evolution of the risk score over consecutive monitoring windows. Reference thresholds for Low Risk and Moderate Risk are plotted to assist in real-time interpretation. Under the classification rule (Low < 0.05 , Moderate $0.05–0.15$, High ≥ 0.15), all 40 windows fall into the "High Risk". The score fluctuates rather than trending steadily upward or downward. The patient recording used for this test set has a persistently elevated presence of abnormal beats. There's no window where the model detects a return to a low-risk state.

3.2.1.7. Feature extraction:

Each Conv1D layer contains a set of learnable filters that operate over the length of the beat waveform and detect localized patterns. The ReLU activation functions after each convolution between waveform shape and beat class, rather than just simple linear separations, by introducing non-linearity. Batch Normalization normalizes the intermediate feature activations. Pooling operations reduce the temporal dimensionality of feature maps, thus lowering computational cost.

3.2.2. Multimodal hybrid model

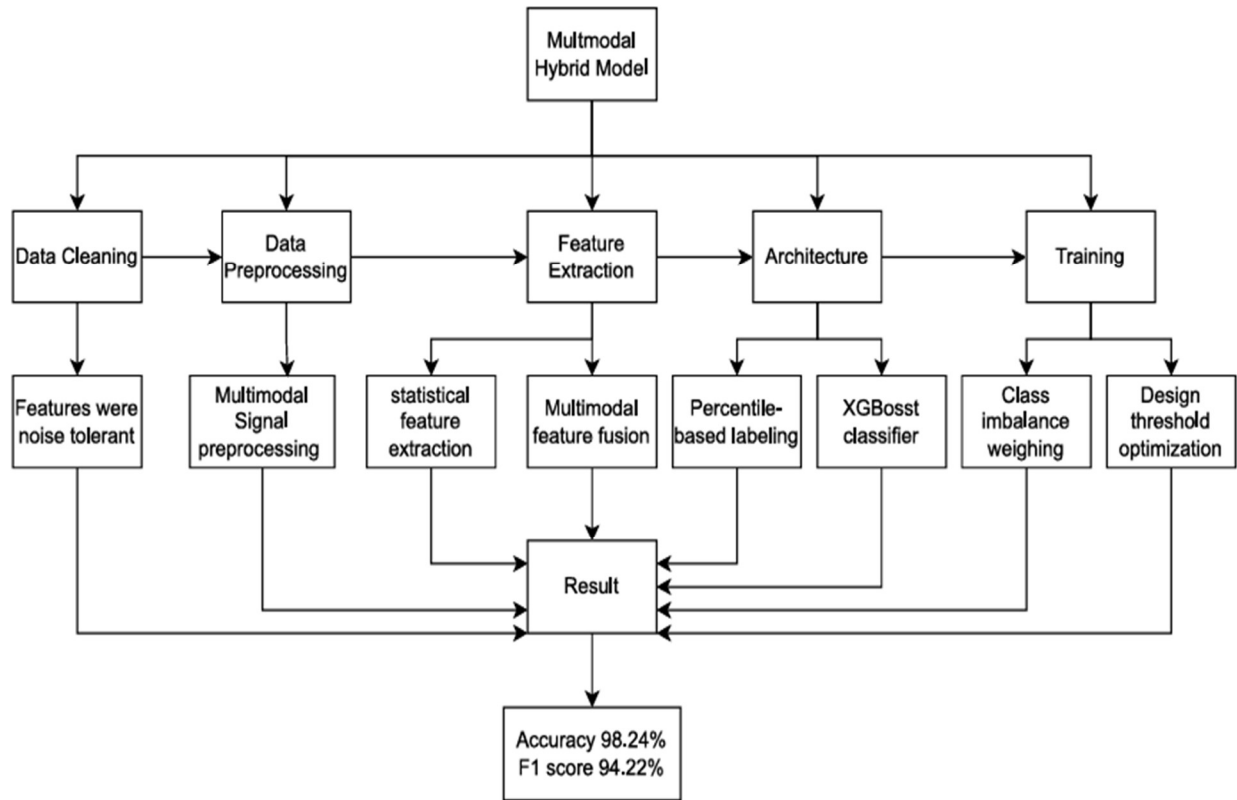


Figure 5: Multimodal hybrid Model

Fig 2 shows that the ECG and PCG data acquired from the Ephnogram database are segmented separately into windows. Then nine statistical features, such as mean, standard deviation, percentiles, and signal energy, are extracted from each type of signal, giving the set of 18 features. The Isolation Forest is trained on these data and gives pseudo-labels based on the anomaly threshold at the 15th percentile. Thus, semi-supervised learning is applied to train the tuned XGBoost model. The probability threshold tuning $t=0.45$ leads to the best F1-score and allows us to get a continuous trend of cardiac risk.

3.2.2.1. Data Cleaning

No data Cleaning was involved in the Multimodal hybrid model as the data received were already clean in the database.

3.2.2.2. Data Preprocessing and Signal Processing

Due to differences in sampling rates and signal characteristics, a structured preprocessing pipeline is applied for both the notebooks:

3.2.2.3. Multimodal Signal Pre-processing, resampled

ECG and PCG signals were pre-processed to ensure temporal and morphological consistency across modalities. Both signal types, originally recorded ere resampled to a common frequency to enable synchronized windowing and joint analysis.

$$X_{resampled} = \text{resample}(X_{signal}, N) \quad (15)$$

where X_{signal} is the raw segmented ECG or PCG window and $N = 200$ is the target number of samples for Pre-processing the signals.

3.2.2.4. Feature Extraction

Statistical Feature Extraction

For each preprocessed signal segment x , a 10-dimensional feature vector is extracted:

$$F(x) = [\mu, \sigma, x_{max}, x_{min}, m, P_{10}, P_{25}, P_{75}, P_{90}, E] \quad (16)$$

where $\mu = \text{mean}(x)$ is the signal mean, $\sigma = \text{std}(x)$ is the standard deviation, X_{max} and X_{min} are the maximum and minimum amplitudes, m is the median, P_{10} , P_{25} , P_{75} , P_{90} are the 10th, 25th, 75th, and 90th percentiles, and E is the signal energy, defined as:

$$E = \sum_{i=1}^N x[i]^2 \quad (17)$$

- $x[i]$ — the amplitude of the signal at sample index i (e.g., a single ECG or PCG voltage reading at a given time point)
- N — total number of samples in the signal
- E — the total energy of the signal, computed by squaring each sample value and summing across all N samples

This has been done in order to get statistically balanced features from the ECG and PCG Signals as part of the feature extraction process.

Multimodal Feature Fusion

The fusion strategy enables the detection of anomalies that may manifest more prominently in one modality than the other, improving the robustness of the anomaly detection pipeline compared to relying on a single signal source,

$$X = [F_{ecg} \mid F_{pcg}] \quad (18)$$

where F_{ecg} and F_{pcg} are the feature vectors extracted from the ECG and PCG channels, and the vector X is formed by concatenating along the feature axis so as to get the fused feature for multimodal variances as part of feature extraction processes.

Training

Class Imbalance Weighting

To address significant class imbalance, present in the dataset, class weights were computed to class frequency and into the training objective. This weighting strategy penalized misclassification of minority classes more heavily. The formulae needed has been given,

$$scale_pos_weight = \frac{N_{negative}}{N_{positive}} \quad (22)$$

where $N_{negative}$ and $N_{positive}$ are the number of normal and anomalous samples in the training set, respectively done in order to calculate any imbalances between so as to balance and get a formidable valuations.

Decision Threshold Optimization

Rather than relying on 0.5 as the decision threshold, classification thresholds were optimized post training by evaluating model performance across a range of threshold activities, ensuing that between sensitivity and specificity was tuned appropriately for the clinical context of arrhythmia and anomaly detection, where false negatives can carry significant risk

$$t^* = \arg \max_t F1(t) \quad (23)$$

$$F1(t) = \frac{2 \cdot Precision(t) \cdot Recall(t)}{Precision(t) + Recall(t)} \quad (24)$$

t^* — the optimal threshold values

arg max_t F1(t) — "the value of t that maximizes F1(t)" — you're scanning across possible threshold values t and picking the one that gives the highest F1 score. This is your threshold-tuning step (e.g., for converting predicted probabilities into class labels).

3.2.2.5. Evaluation

These are the formulas that at the end to end has been done in order to calculate our variation accuracy, F1 and other valuations score metrics needed to be calculated in order to be able to compare the other metrics research papers with ours.

$$Accuracy = \frac{TP+T}{TP+TN+FP+FN} \quad (25)$$

$$Precision = \frac{TP}{TP+F} \quad (26)$$

$$Recall = \frac{TP}{TP+F} \quad (27)$$

$$F1 = \frac{2 \cdot Precision \cdot Recall}{Precision + Rec} \quad (28)$$

- **TP (True Positive)** - model correctly predicted the positive class (e.g., correctly flagged an arrhythmia/anomaly)
- **TN (True Negative)** - model correctly predicted the negative class (correctly identified normal/healthy)
- **FP (False Positive)** - model incorrectly predicted positive
- **FN (False Negative)** = model incorrectly predicted negative (missed an actual abnormal case)

These are the formulas that at the end to end has been done in order to calculate our variation accuracy, F1 and other valuations score metrics needed to be calculated in order to be able to compare the other metrics research papers with ours.

4. RESULTS & DISCUSSION

When marked against recent work in AI driven cardiac and health monitoring, the proposed system reflects both the strengths and the open challenges identified across the literature. Sharma et al [1] and Ramzan et al [2] report strong classification accuracies (92–95% and 96.8%, respectively) for general physiological anomaly detection using classical ML models such as ANN, SVM, and Random Forest, but both rely on synthetic or aggregated IoT datasets rather than clinically sourced, subject specific waveform data, limiting direct comparability with the beat-level ECG classification performed in this work. In contrast, the 1D-CNN arrhythmia classifier developed here, evaluated on the MIT BIH database, achieved a macro F1 score of 0.81 across four arrhythmia classes after addressing class imbalance via SMOTE a result broadly consistent with, though more conservative than, the near-perfect accuracies reported by deep learning-based ECG classifiers such as MCA-net W.Pan et al [9] and the patch-lead CNN approach of Lai et al. [8], both of which benefit from larger, more balanced 12 lead datasets (PTB/PTBXL and 55 patient Holter recordings, respectively) compared to the single lead, imbalanced setting addressed in this study. For the multimodal component, the present system's use of IF derived pseudo-labels combined with XGBoost classification addresses a labeled-data problem similar to that discussed by Soumma et al. [4] and Huang et al. [5], but takes a semi-supervised route rather than assuming access to expert-annotated ECG-PCG pairs, as in Li et al.'s [10] multimodal CNN-LSTM framework, which reported an AUC of 0.936 using fully labeled PhysioNet/CinC data. This distinction is significant: while Li et al. [10] demonstrate that multimodal fusion of ECG and PCG deep-coded features substantially outperforms single-modality models (ECG-only AUC of 0.916 versus PCG-only 0.748), the present work cannot claim equivalent validated performance, since its "ground truth" for the multimodal task is derived from unsupervised anomaly detection.

4.1. Distribution Visualization

To understand the impact of class imbalance on model training, the class distribution of the arrhythmia dataset is visualized before and after applying SMOTE. Figure 6 illustrates this comparison through pie charts, highlighting the original imbalance in the MIT-BIH dataset and the balanced distribution achieved in the oversampled training set. Subsequent visualizations, including the confusion matrices, further illustrate how this rebalancing translates into per-class model performance.

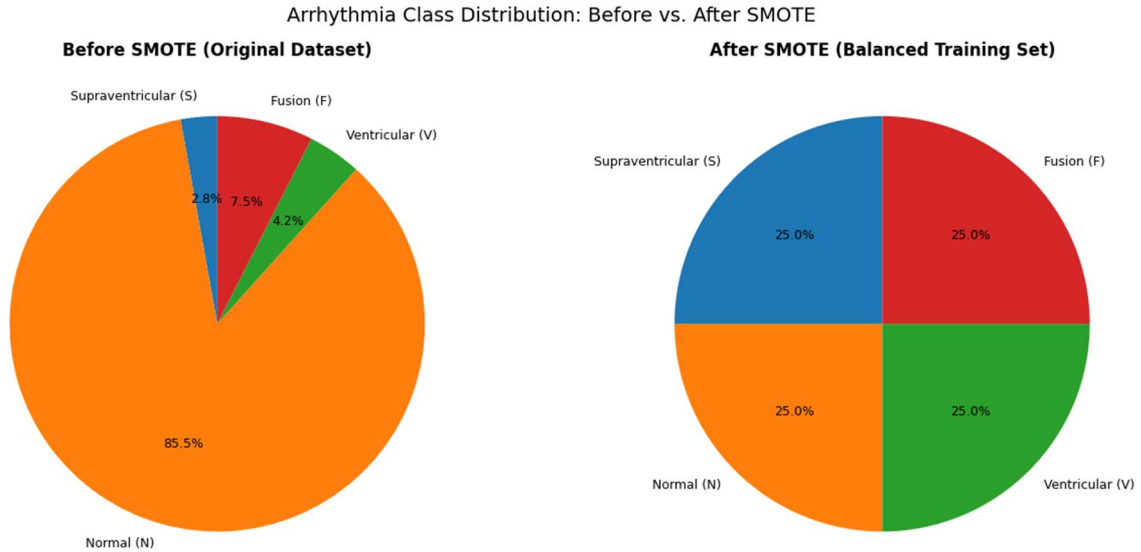


Figure 6: Pie chart visualization

The figure 6 shows that the MIT-BIH Arrhythmia Database exhibits class imbalance, with N beats constituting the majority of the dataset, while the minority classes S, V, and F are underrepresented samples. The Synthetic Minority Oversampling Technique, SMOTE, has been applied to training set generating synthetic samples for each minority class between existing minority class and nearest neighbors until all four classes were equally represented. SMOTE

was applied to the training data, while the test set retained its original distribution to ensure reported performance metrics remain free from data leakage introduced by oversampling.

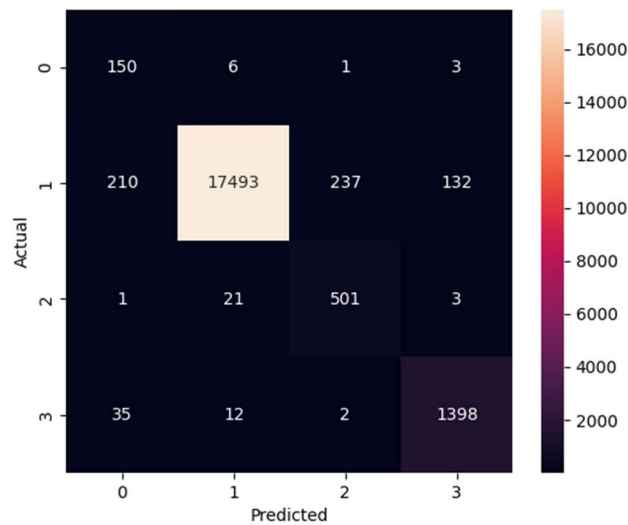


Figure 7: Confusion matrix -f 1D CNNs beat classification

Figure 7 is the confusion matrix for the 1D CNN's beat classification on the test set. Under the classification rule , all 40 windows fall into the "High Risk" category. The consistently high, flat trend suggests the patient recording used for this test set has a elevated presence of abnormal beats throughout not just in isolated moments.

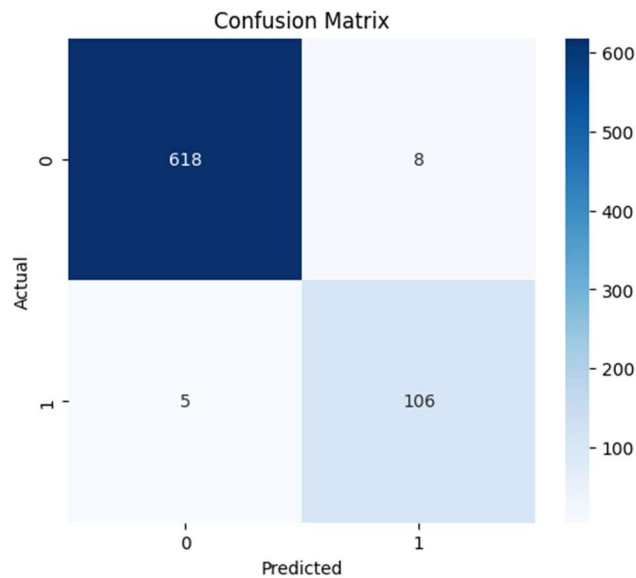


Figure 8: Confusion matrix Multimodal Isolation Forest + XGBoost anomaly detector.

Figure 8 shows the confusion matrix for the multimodal anomaly detection pipeline on the test set, where class 0 represents normal cardiac windows and class 1 represents detected anomalies. Out of 626 true normal instances, 618 were correctly classified while only 8 were misclassified as anomalies, giving a low false-positive rate. The 111 true anomalous instances, 106 were correctly identified, with only 5 missed. This strong diagonal concentration confirms the high accuracy (98.24%), precision (92.98%), and recall (95.50%) ,and indicates that the Isolation Forest XGBoost

pipeline distinguishes anomalous cardiac windows from normal ones, with a bias toward minimizing missed anomalies over avoiding false alarms .

Table 2. *Per-Class Classification Report (Test Set)*

Class	Precision	Recall	F1-Score	Support
Fusion (F)	0.38	0.94	0.54	160
Normal (N)	1.00	0.97	0.98	18,072
Ventricular (V)	0.68	0.95	0.79	526
Supraventricular (S)	0.91	0.97	0.94	1,447
Accuracy			0.97	20,205
Macro Avg	0.74	0.96	0.81	20,205
Weighted Avg	0.98	0.97	0.97	20,205

The N class, which dominates the test set, achieves performance (precision 1.00, recall 0.97, F1-score 0.98), as expected given its representation. The S class also performs strongly (F1-score 0.94) despite a smaller portion of the data samples even under imbalance. The V class shows a more moderate result (F1-score 0.79), with high recall (0.95) but lower precision (0.68), indicating the model identifies most true V beats but produces false positives from other classes. The F class, the smallest and most underrepresented at only 160 samples, shows the weakest performance (F1-score 0.54): although recall is high (0.94), meaning the model rarely misses, precision is low (0.38), meaning the majority of beats predicted as Fusion are actually misclassifications from other classes. This precision-recall imbalance in the minority classes is consistent with the class imbalance, and reflects why SMOTE based oversampling. The overall accuracy of 0.97 is driven largely by the N class; the macro average F1 score of 0.81 provides a more honest measure of the model's balanced performance across classes and is lower than the weighted average (0.97), which is skewed by the Normal class.

Table 3. *Multimodal Anomaly Detection Performance*

Metric	Value
Accuracy	98.24%
Precision	92.98%
Recall	95.50%
F1 Score	94.22%

The model achieves an overall accuracy of 98.24%, with a precision of 92.98%, recall of 95.50%, and an F1-score of 94.22%. The high recall relative to precision indicates that the system is slightly more effective at capturing true anomalies than at avoiding false positives, which is a favourable trade-off in a clinical screening context where missing an anomalous cardiac event carries a higher cost than flagging a borderline case for further review. The

strong F1-score suggests that the fusion of ECG and PCG features through the XGBoost classifier is effective at distinguishing anomalous from normal cardiac windows.

4.2. STATE OF THE ART COMPARISION

Table 4. *Performance comparison with other studies*

Author(s)	Databases	Algorithms	Evaluation
Z. Yan, J. Zhou et al. [7]	MIT-BIH	CNNs	ACC:90%
D. Lai, Y. Bu et al. [8]	MIT-BIH	CNNs	ACC:93.1
W. Pan, Y. An et al. [9]	PTB-PTBXL	CNNs	Acc: 88.59
P. Li, Y. Hu, and Z. P. Liu et al. [10]	PhysioNet/CinC	GA-SVM	Acc: 87.40
Our Method	MIT-BIH/ Ephnogram	CNNs / Isolation Forest - XGBoost	Acc: 98.24 F1 Score – 94.22

Yan et al [7] employed a CNN-based architecture and reported an accuracy of 90%, while Lai et al. [8] achieved a slightly higher accuracy of 93.1% using a CNN model combined with R-R interval features on patch-based single-lead ECG recordings. Pan et al. [9], despite using a more sophisticated multi-task channel attention CNN architecture (MCA-net), reported a comparatively lower accuracy of 88.59% on the detection task, while Li et al. [20], who adopted a hybrid GA-SVM approach that fused deep-coded ECG and PCG features via genetic algorithm-based feature selection, achieved 87.40% accuracy. In comparison, the proposed method which combines a 1D-CNN for ECG arrhythmia classification with an Isolation Forest-XGBoost pipeline for multimodal anomaly detection achieves the highest accuracy among the compared approaches at 98.24%, along with an F1-score of 94.22%. This improvement suggests that combining deep learning-based feature extraction with an ensemble-based classifier and unsupervised pseudo-labelling can offer a competitive and in this comparison superior alternative to purely CNN-based or single-classifier approaches for cardiac signal analysis.

CONCLUSION

A 1D CNN trained on the MIT BIH dataset, classified heartbeats in four AAMI classes with 97% accuracy score and a macro F1 score of 0.81 and was able to effectively address class imbalance with weighted training. In the second method, the ECG and PCG signals were fused by combining Isolation Forest pseudo-labelling and a tuned XGBoost classifier, achieving 98.24% accuracy score and 94.22% F1 score, indicating the advantage of multimodal fusion. With the referred papers having a better accuracy and F1 score the research papers project has been maximized to the current needed evaluation of maximizing its accuracy and f1 score to the fullest.

FUTURE WORK

If any such Future work, focus on validating the multimodal anomaly detection pipeline against annotated ECG-PCG data rather than Isolation Forest-derived labels, and evaluation to the full Ephnogram database to improve. Further improvements to minority-class performance (particularly the Fusion class) could be explored through advanced imbalance-handling techniques such as focal loss or cost-sensitive learning. Additionally, re-evaluating the proposed method on standardized benchmarks (e.g., MIT-BIH, PTB/PTBXL) would enable more comparison with existing methods. Finally, incorporating explainable AI techniques such as SHAP or Grad-CAM would improve model interpretability and support clinical adoption.

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